S4 Table: Benchmark Expenditure and Operational Audit

Candidate number: 1102384

All agentic workflows were capped at 50 messages and 500,000 cumulative tokens. Table S4 retains all-attempt resource and operational accounting.

Table S4: Benchmark expenditure and operational audit. Panel A separates the fixed target-cell denominator from persisted attempts and partitions provider-reported cost into candidate-model, search and grading charges. Monetary components and totals in Panel A are independently rounded. "Extra" is Attempts minus Targets and includes failed, recovery and retry attempts. Panel B reports outcomes for the three analysed agentic workflow submissions per model; all 24 analysed single-response submissions failed clean execution and are omitted from that panel. Executed denotes a clean zero-exit replay; Complete denotes all required non-empty files; Exact replay denotes reproduction of the designated numerical outputs; All three requires every condition. Retry denotes a selected retry. Costs are US dollars at the execution dates and exclude local computation, storage, human time and unexported charges.

Panel A: all-attempt accounting

Task class	Configuration	Targets	Attempts	Extra	Model	Search	Grading	Total
Conceptual	Single response	400	408	8	1.614	0.000	6.342	7.956
Conceptual	Workflow	400	402	2	5.329	1.096	7.689	14.114
Calculation	Single response	40	44	4	1.625	0.000	1.072	2.698
Calculation	Workflow	40	40	0	0.827	0.000	0.832	1.659
Extended modelling	Single response	24	24	0	1.638	0.000	0.930	2.568
Extended modelling	Workflow	24	41	17	2.153	0.380	2.958	5.491

Panel B: analysed extended-task agentic workflow outcomes

Model	Retry	Executed	Complete	Exact replay	All three
GPT-5.6 Sol	2	3/3	3/3	3/3	3/3
Claude Opus 5	1	3/3	2/3	2/3	2/3
Gemini 3.6 Flash	3	2/3	2/3	2/3	2/3
DeepSeek V4 Pro	3	2/3	1/3	2/3	1/3
GPT-5.4 Mini	2	3/3	3/3	3/3	3/3
DeepSeek R1	0	1/3	1/3	1/3	1/3
Qwen 3.5 9B	3	1/3	0/3	0/3	0/3
Ministral 3B	3	1/3	1/3	0/3	0/3